

The RUSH Checklist: A Standardized Framework for Reporting User Studies in Human-Robot Interaction

Shruti Chandra*

University of Northern British Columbia
Prince George BC Canada
shruti.chandra@unbc.ca

Katie Seaborn

Institute of Science Tokyo
Tokyo Japan
University of Cambridge
Cambridge UK
katie.seaborn@cst.cam.ac.uk

Giulia Barbareschi

Keio University
Yokohama Japan
University of Duisburg-Essen
Duisburg Germany
giulia.barbareschi@uni-due.de

Wing-Yue Geoffrey Louie

Oakland University
Rochester MI USA
louie@oakland.edu

Shelly Bagchi†

U.S. National Institute of Standards and
Technology
Gaithersburg MD USA
shelly.bagchi@nist.gov

Sara Cooper

Artificial Intelligence Institute - CSIC
Barcelona Spain
sara.cooper@iia.csic.es

Zhao Han

University of South Florida
Tampa, FL USA
zhaohan@usf.edu

Daniel Tozadore

University College London
London UK
d.tozadore@ucl.ac.uk

Abstract

Transparent and consistent reporting of user studies is essential for advancing scientific knowledge. In human-robot interaction (HRI), studies are often reported incompletely, even in top-tier venues, limiting proper evaluation, replication, and practical application of findings in practice. This study aimed to generate expert consensus on a reporting checklist for HRI user studies and provide a validated tool to improve transparency, reproducibility, and methodological rigor in the field, leading to easier translation of research into practice. A two-round Delphi study was conducted with 34 HRI experts from academia and industry from over 12 countries. An international panel of nine interdisciplinary experts first synthesized a preliminary list of 116 reporting items from the literature. Experts rated the importance of each item and provided qualitative feedback. Consensus was defined as 70% agreement, and items were iteratively refined through anonymous online surveys. Overall, consensus was achieved on 106 items, encompassing both essential and context-dependent elements in nine domains. The resulting RUSH checklist (Reporting User Studies in Human-Robot Interaction) provides the first community-endorsed, consensus-based reporting guideline for HRI user studies.

*Corresponding author: shruti.chandra@unbc.ca

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CCS Concepts

• **Computer systems organization** → **Robotics**; • **Human-centered computing** → **User studies**; • **Information systems** → **Data replication tools**.

Keywords

Human-Robot Interaction, Reporting Standards, Reporting Template, Replicability, Reproducibility, Guidelines, Delphi Study

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1 Introduction

Human-Robot Interaction (HRI) as a discipline is grounded in empirical user studies, which provide the essential evidence base for understanding how people perceive, engage with, and benefit from robotic systems across diverse contexts [5, 16]. As the field expands, the quality, completeness, and transparency of study reporting play a critical role in shaping the reproducibility [14] of individual research contributions as well as the cumulative HRI knowledge [22]. However, recent reviews highlight persistent shortcomings in reporting practices across both top-tier venues and other publication outlets [9, 38, 40]. A recurring concern is that many HRI papers omit fundamental details, including participant demographics [38], recruitment strategies [9], compensation structures [9], statistical reporting [22], methodological descriptions, and metrics used [43]. Such omissions hinder not only replication but also meaningful comparison and critical evaluation of findings. These shortcomings have prompted a call to action [22, 40] and motivated the development of community guidelines and best practices [4, 7, 41].

Despite ongoing efforts, HRI still lacks a reporting framework that reflects its interdisciplinary nature. Other fields facing similar challenges have established standardized guidelines to improve rigor and transparency. In medicine, CONSORT (Consolidated Standards of Reporting Trials) became the gold standard for clinical trials [25], while PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) [30] brought consistency to systematic reviews. Psychology strengthened reproducibility through the APA Journal Article Reporting Standards (JARS) [3, 32], and qualitative research adopted SRQR (Standards for Reporting Qualitative Research) to ensure rigor and comparability [29]. The EQUATOR (Enhancing the Quality and Transparency Of health Research) Network further integrated diverse guidelines, such as CERT (Consensus on Exercise Reporting Template) [39], demonstrating how reporting standards can be adapted across domains.

Recently, Bagchi et al. [4] recognized a parallel need in HRI and proposed recommendations for reporting human studies, including structured checklists and appendices. However, validation and refinement through broad community consensus were not carried out, and this may lead to challenges in adopting a reporting standard, as prior research often was unable to reach consensus on terminology, relevant contextual variables to a study, and methods of evaluation [10, 38, 43]. Given the unique challenges that HRI research presents, including in robot embodiments, interaction contexts, and its interdisciplinary scope, there is a pressing need for a reporting standard grounded in expert consensus to ensure that proposed tools align with community research practice and are thus more likely to be applied by scholars in the field. Achieving this requires systematic engagement with diverse stakeholders in the HRI community.

To address this gap, we conducted a Delphi study [1, 8, 13, 27, 28] to generate expert consensus on a reporting checklist for HRI user studies. Global experts, including professors, researchers, and industry professionals, were invited to anonymously and asynchronously review and refine a synthesized checklist that the authors initially derived from literature. Feedback from each round led to iterative revisions and convergence on a consensus. The outcome was a validated reporting standard. Our main contributions include:

- **RUSH (Reporting User Studies in Human-Robot Interaction), a community-endorsed reporting checklist for HRI studies, developed through global expert consensus:** We created the first validated, community consensus-based reporting guideline for HRI user studies by engaging 34 global HRI professionals in a two-round Delphi study.
- **A method of flexibly and comprehensively reporting on HRI study domains:** The checklist addresses both essential and context-dependent elements across all critical aspects of HRI user studies in nine domains including study design, participants, ethical conduct, data collection and analysis, results and discussion, and code and data release, comprising 106 reporting items to ensure a thorough and standardized reporting approach.
- **A proof-of-concept approach to developing consensus and standards leveraging Delphi study methodology.** We demonstrate the successful application of the domain-neutral Delphi study methodology to HRI practice. This sets the stage for other consensus-based initiatives in HRI.

We hope to make the RUSH checklist an official standard or informal norm. The first step is uptake, feedback, and endorsement by the wider community. Towards this end, we will provide a hub at <https://aspirelab.io/rush>. We expect to iteratively refine the framework after feedback cycles and actual use. The next step will be to apply for standard use at venues like the ACM/IEEE HRI conference in the hope that the HRI community will join us in these efforts to standardize and bolster research practice in our field.

2 Background

HRI researchers have repeatedly shown that current reporting practices are inconsistent and often incomplete, raising critical concerns about reproducibility [9]. The lack of details encompasses nearly all aspects of reporting, from participant metadata to methods, outcomes, and ethical considerations. A meta-analysis of 416 HRI studies published between 2019 and 2021 [9] revealed substantial gaps in reporting study metadata. Information on participant recruitment, compensation, and gender was frequently omitted; for instance, 259 (62.3%) studies failed to report recruitment methods and 257 (61.8%) did not mention compensation.

Representation in participant populations presents another area of concern. A systematic review of 827 HRI studies published in the ACM/IEEE HRI conference proceedings (2006–2022) by Seaborn et al. [38] found pervasive WEIRD (Western, Educated, Industrialized, Rich, Democratic) sampling bias, along with significant gaps in reporting factors such as race, age, and experience levels. Their analysis further noted that many studies omit essential participant details, rely on non-standard metrics (e.g., median ages), or use vague descriptors (e.g., “approximately 100 participants”), thereby limiting interpretability and hindering meta-analyses. Similar inconsistencies were highlighted in reporting methods and outcomes. Zimmerman et al. [43] analyzed HRI conference papers (2015–2021) and found insufficient transparency and standardization in the measurement and reporting of interaction outcomes, complicating cross-study comparisons. Riek [34] examined Wizard-of-Oz studies in HRI, uncovering inconsistencies in experimental setups and proposing guidelines to improve methodological rigor.

The consequences of such omissions are evident in replication challenges. Gunes et al. [14] identified critical obstacles to reproducibility stemming from the complex interplay of social, cognitive, and technological factors in HRI studies, underscoring the need for more comprehensive and transparent reporting practices. Incomplete reporting also limits the ability to perform meta-analyses or aggregate findings across studies, slowing the accumulation of generalizable knowledge and the development of evidence-based HRI guidelines. Moreover, lack of consistency in what is reported hinders efforts to aggregate and disaggregate collective knowledge; it is harder to build cross-sectional and longitudinal knowledge from multiple studies [22, 40]. In addition, ethical reporting remains a largely neglected dimension of HRI publications. Rosén et al. [36] analyzed 73 papers from ACM/IEEE HRI, RO-MAN, and THRI (2018–2019) and found that four of five core principles of ethical conduct: ethical approval, data protection, deception, and debriefing, were reported in less than a third of studies, while informed consent appeared in only 49%. To begin addressing some of the shortcomings highlighted in these reviews, it is important to

understand what the existing guidelines for reporting in HRI are, what efforts have already been made to improve them, and unpack the reasons that might have affected their uptake thus far.

2.1 Proposed Reporting Frameworks in HRI

Recognizing the existing challenges, researchers have proposed structured approaches to improve reporting in HRI. Baxter et al. [6] analyzed HRI studies from 2013 to 2015, revealing inconsistencies in experimental design, measurement, and participant sampling, and emphasized the need for standardized reporting of procedures, tasks, and metrics. They proposed six key recommendations: R1: Motivation and Goals, R2: Robot Autonomy, R3: Ecologically Valid Subjects and Environments, R4: Long-term Interactions and Novelty Effects, R5: Descriptive Statistics, and R6: Replication and Reproducibility. The recommendations serve two purposes: at the researcher level, to provide a systematic standard for planning, conducting, analyzing, and reporting studies; and at the field level, to promote collaborative tools, standardized reporting, and infrastructure to enhance transparency and reproducibility in HRI. Tavares et al. [41] identified 26 essential characteristics for reproducible studies across five categories: 1) Description, 2) Design of Experiment, 3) Procedures, 4) Sampling, and 5) Transparency by reviewing prior work. They also introduced a quantitative ‘Reproducibility Score’ to assess reporting completeness, replicability, and reliability.

Leichtmann et al. [22] attribute the replicability crisis to factors such as theory, measurement, statistics, reporting, and scientific incentives. They compare HRI practices with other fields, review meta-analyses, and summarize replication attempts, identifying issues such as small sample sizes, weak theoretical foundations, and incomplete reporting. They recommend statistical, methodological, and social reforms to improve rigor and reliability, but fall short of an operational framework for implementation. To address this gap, Bagchi et al. [4] introduced preliminary human-centric reporting checklists based on established practices in Human-Computer Interaction (HCI) and psychology. Building on this, Holthaus et al. [17] developed a web-based reporting form to improve study comparability and replicability. A formative evaluation with nine experts highlighted its value for authors, reviewers, and the broader community, while suggesting refinements in format, clarity, and qualitative reporting. This work marks an important step toward embedding formalized reporting practices into the HRI scholarly ecosystem, though broad community uptake is unclear. Ultimately, the success of such initiatives to improve practice is not only determined by the quality of the tool but also by community engagement. It is therefore valuable to examine how practitioners in other fields have engaged their communities in building standardized tools that fit everyday research practices.

2.2 Lessons from Other Disciplines: The Need for Community-Based Standards

HRI is an interdisciplinary field that brings together insights from computer science, engineering, psychology, sociology, ethics, and more. Such a plurality enriches the depth and breadth of the research, but can also make standardizing practice extremely challenging, as tools and methods need to account for the differences across the spectrum [42]. Building effective reporting standards

thus requires collaboration, broad support, and sustained community engagement, rather than prescriptive top-down mandates.

Reporting guidelines across disciplines have consistently emerged from rigorous, community-driven development processes, rather than top-down mandates. For instance, CONSORT [25], the gold standard for clinical trial reporting, was developed through international workshops and refined via multiple rounds involving statisticians, editors, and stakeholders. Similarly, PRISMA [30] evolved from the “Quality of Reporting of Meta-Analyses” statement and was shaped through extensive consensus-building workshops, broad stakeholder engagement, and iterative public consultation. Extensions of these guidelines continue to follow consensus meeting methodologies, ensuring inclusivity across fields. The EQUATOR Network [39] institutionalizes such processes, providing collaborative forums for guideline developers and researchers to co-create and refine reporting standards, such as the CERT checklist from Gattrell et al. [12], through structured consensus processes. Across these cases, community-driven, iterative, and inclusive processes have yielded reporting frameworks that balance flexibility with rigour and achieved broad trust and adoption. Comparable movements are evident in computing-related fields. In HCI, reproducibility and reporting have become active topics of debate, with initiatives such as artifact evaluation and reproducibility checklists being adopted in major conferences [33]. In Artificial Intelligence (AI) and machine learning, venues like NeurIPS (Neural Information Processing Systems)¹ have introduced reproducibility programs to promote discussion and provide tools for standardizing reporting practices [31]. Within the HRI community, calls for stronger reporting standards are already emerging [4, 9, 17, 38].

Drawing from this, the authors of this paper advocate for the development of community-based reporting standards in HRI. Such standards would encompass ethical considerations, experimental design, data reporting, and performance metrics, ensuring that HRI studies are conducted with rigor and transparency.

3 Methodology: Delphi Process & Consensus-Building

We employed the Delphi method, a structured, multi-phase process for collecting anonymous feedback and building consensus iteratively, reducing the influence of dominant voices and enabling a balanced perspective across geographic and disciplinary lines [13, 28].

This method has been widely adopted to develop cohesive guidelines and inform best practice. Delphi studies have been used in health research to design and validate reporting checklists, such as the “Consensus on Exercise Reporting Template” [39], and to standardize procedures for usability evaluations of digital health solutions [23]. In clinical practice, the method has guided consensus on skill recommendations for brain ultrasonography in critical care [21, 35]. In technology, the method was used to elicit expert knowledge on emerging innovations [1], higher education [8], and academic programs and teaching practices [20]. The Delphi method is versatile as a structured, iterative, and anonymous process for building expert consensus in contexts where standards or evidence are limited. In our study, experts completed two online surveys

¹NeurIPS is a leading conference in machine learning and computational neuroscience and artificial intelligence research.

designed to capture and refine essential reporting elements over two rounds. The second round built on the previous one to generate deeper and more refined feedback. Our aim was to create a community-informed reporting checklist to strengthen transparency, reproducibility, and methodological rigor in HRI research.

3.1 Author Backgrounds

To ensure rigor, relevance, and inclusivity in the study design and analysis, the research was conducted by an interdisciplinary team of authors with diverse backgrounds in HRI, spanning academic, governmental, and industrial contexts. Each of us brought expertise in both methodological approaches and domain-specific knowledge. Our collective backgrounds encompass social robotics, healthcare and assistive technologies, child-robot interaction, industrial and manufacturing robotics, educational technologies, and real-world robot deployments. Methodologically, the team is experienced in mixed methods, experimental design, co-design, statistical and content analysis, and systematic reviews. For instance, SC1 (Canada), an assistant professor, focuses on real-world applications of social robots for education and healthcare; KS (Japan/Canada/UK) is an associate professor with expertise in mixed methods and systematic reviews; GB (Germany/Japan) is a professor with experience working on accessibility and inclusive technology research; SB (USA) is a federal researcher specializing in industrial robotics, standards, and metrics; GL (USA), an associate professor, focuses on teachable robots and HRI for education, healthcare, and security applications; SC2 (Japan/Spain) works as a social robotics researcher in industry; DT (UK/Switzerland) is a lecturer working on AI in multi-modal education and stakeholder-driven design; and ZH (USA), an assistant professor, works on AI-powered transparent robots in mobile manipulation and persuasive social robots.

3.2 Participants

Participants were sought from our global networks: active researchers in the field of HRI, including professors, research assistants, postdoctoral researchers, or scientists working in industry or government agencies with a minimum of five years of experience in the HRI field and a minimum age of 18. 34 experts aged 27–69 years ($n = 34$, $M = 41.9$, $SD = 10.5$) participated, 18 identified as men, 16 as women, and none of another gender. Participants were based in: the United Kingdom (6), Germany (5), the United States (4), Canada (2), Portugal (2), Japan (2), Sweden (1), Switzerland (1), Italy (1), France (1), Australia (1), Türkiye (1), multiple countries (Canada/Japan) (1), and unspecified (6). Ethnic identities (self-reported; multiple responses allowed) included: White, European and Caucasian (25), Asian (3), East Asian (1), Japanese (1), Latin American (1), and 5 participants preferred not to disclose. Most participants held a Ph.D. (33), with two others holding advanced degrees (Doctor of Science) (1), Doctor of Engineering (1). One participant was a Ph.D. student. Participants reported working in various roles, including: Full Professors (10), Associate Professors (5), Assistant Professors (9), Researchers in Academia (9; overlapping with faculty roles), Researchers in Industry (3), and a Ph.D. Student.

In terms of their experience in HRI, participants $n = 21$ reported an average of 5.6 years of experience as students in HRI-relevant fields and 10.2 years as researchers or faculty, excluding student

years. $n = 34$ had disciplinary backgrounds in Computer Science (13), Robotics (16), Psychology (12), Engineering (10), Design (5), Humanities (1), Sociology (1), Gender Studies (1), Bioengineering (1), Assistive Robotics for people with special needs (1), and Transdisciplinary Studies (1). Participants were motivated to contribute due to professional interest, personal invitations, or a desire to support the HRI research community.

3.3 Ethics

The study was reviewed and approved by the Ethics Committee of Institute of Science Tokyo (formerly Tokyo Institute of Technology; approval #2023263). All potential participants received details about the study and provided informed consent online. Notably, all were informed that their data would be collected anonymously, with no personally identifiable information recorded.

3.4 Survey Tool

We used Qualtrics, a secure web-based platform, to administer the survey. The Round 1 survey had 116 items (Consensus Report Appendix A), which were developed and iteratively refined by the research team to ensure clarity, relevance, and alignment with the project objectives. The items were grouped into the following domains, with the corresponding number of items in each: General Information (11 items), Introduction and Background (8), Participants (33), Study Design (24), Ethical Conduct (6), Data Collection (18), Results (6), Discussion (5), and Code and Data Release (5).

3.5 Procedure and A Priori Decisions

We used an asynchronous online questionnaire. The survey included study information outlining the procedure, a consent form, and the main questionnaire. The questionnaire consisted of two parts: first, participants provided their background and demographic information; second, they rated and gave feedback on a set of items related to reporting HRI user studies. Experts rated each item on a 7-point Likert scale and could provide open-ended comments for each item. All responses were collected anonymously.

The study employed a two-round Delphi approach. In the first round, experts completed the initial questionnaire, which took approximately 30–40 minutes, with instructions emphasizing that they should focus on the inclusion, exclusion, and definition of each item, without commenting on the structure or order of the questionnaire, and that items marked as [if applicable] referred to situations when an item was relevant to a particular study. Items that reached consensus for either inclusion in the checklist or exclusion from it were removed before the second round. This approach was used to ensure that the second round would only focus on contentious elements that required further discussion. Based on participant feedback, some items were revised, reformatted, or new questions were added for the follow-up round. The same experts were invited to participate in the second round, which also lasted about 30–40 minutes. The results of the first round were also sent to the participants, along with the survey for the second round. In both rounds, participants were given 3–4 weeks to complete the survey. Weekly reminders were sent to participants to maximize completion. For the second round, an additional instruction was introduced clarifying that items marked as [After peer review]

would only be considered in the final version of the article, and the wording of [if applicable] was refined to [Only if applicable to the study]. This iterative process of two rounds allowed refinement of the item list based on collective expert input.

A priori consensus for including an item in the checklist was defined as greater than 70% of respondents rating 0 or 1 on the 7 Likert scale (0 = strongly agree to 6 = strongly disagree). A priori consensus for removing an item from the checklist was defined as greater than 70% of respondents rating an item as 5 or 6 on the 7-point Likert scale (0 = strongly agree to 6 = strongly disagree).

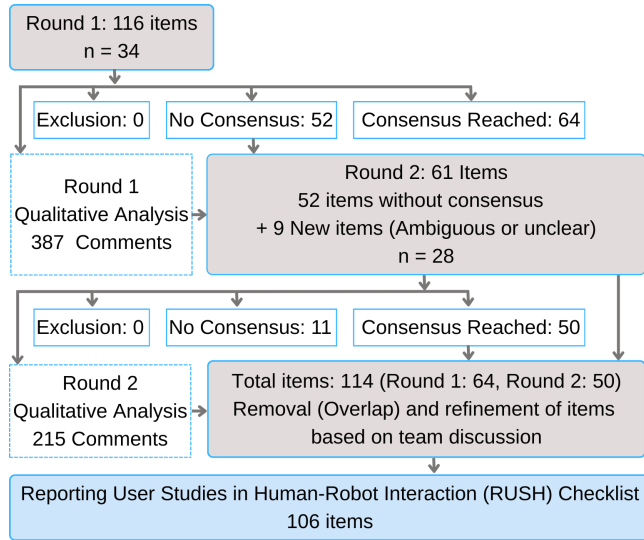


Figure 1: Flowchart on consensus on RUSH items through the Delphi Study. n: number of participants.

4 Analysis

Data from both rounds were analyzed to evaluate consensus and interpret feedback. For each item, consensus was assessed as described in the previous section. In Round 1, each item was rated on a 7-point Likert scale, ranging from 0 (strongly agree) to 6 (strongly disagree). We measured the agreement score by combining the ratings for “strongly agree” and “agree”, as previously suggested in similar studies to streamline analysis and accommodate for nuances in reasoning for strength of agreement [19, 24]. In Round 2, items were rated on a 6-point Likert scale (0: strongly agree to 5: strongly disagree). As our study included two rounds of Delphi, in the second round we removed the “neither agree nor disagree” option, a strategy previously employed by Froud et al. [11] to encourage more decisive responses and facilitate interpretation of agreement or disagreement. The agreement score combined the ratings of “strongly agree”, “agree”, and “somewhat agree.”

For qualitative analysis in Round 1, two coders independently reviewed the comments on non-consensus items, discussed the identified codes, and, based on their collective insights, refined item wording, merged or split items, added new items where needed, and revised the survey for Round 2. In Round 2, to ensure the participants’ comments were fully incorporated into the final checklist,

the team developed a coding scheme, and eight coders independently coded the comments into categories such as suggestions, support, confusion, and requests for clarification. The team then discussed and resolved any disagreements to finalise the coding. Coding frequencies and consensus metrics were tracked across rounds to evaluate convergence of expert opinion and assess the clarity and applicability of the final checklist items.

5 Findings

Fig. 1 summarizes the results of individual rounds and the item consensus flow. We now present our findings by Delphi round.

5.1 Results of the Delphi Process: Round 1

Thirty-four HRI experts participated in Round 1, evaluating 116 items using a 7-point Likert scale (0: strongly agree to 6: strongly disagree). Based on the a priori criterion, participants reached consensus on including 64 items (out of 116, 55%) in the checklist. The remaining 52 items did not meet sufficient consensus and were re-evaluated in the second round. No item reached consensus for exclusion (70% threshold of experts voting 5 or 6 on the Likert scale, common threshold used in Delphi studies [15, 37]).

In total, 387 comments were received in Round 1; Consensus Reached: 191 (49%); No Consensus: 196 (51%). Two authors independently reviewed and discussed 196 non-consensus comments until consensus. Comments addressed issues like adding “Only if applicable to the study” to some items and rewording explanations; notably, some participants overlooked the existing applicability notes, so these were not revised further. Nine new questions were added to Round 2, and one question was modified. Of these, five items were created by splitting up the original abstract item: Background, Purpose, Method, Results, and Conclusion. Two additional questions asked whether contributions and significance, and motivation and long-term goals, should be combined under one heading. Another item was added regarding dropouts: whether related items should be merged. Finally, one new item was included under study context: Household Composition (e.g., roommates, dogs, children, caretakers). This resulted in 61 items for Round 2: 52 carried over from Round 1 and nine new or modified items.

5.2 Results of the Delphi Process: Round 2

Round 2 involved 28 HRI experts (82% retained), and they refined the set of 61 items. The Likert scale was modified from 7 to a 6-point version (0: strongly agree to 5: strongly disagree), by removing the “neither agree nor disagree” option for more decisive responses. Using the same >70% agreement threshold for inclusion, consensus was reached on 50 items (82%). The remaining 11 items did not meet the consensus criteria, and as in Round 1, no items were excluded.

Participants provided 215 comments in Round 2, with consensus reached on 151 (70.23%) and no consensus on 64 (29.7%). Compared to Round 1, the volume of comments decreased by 45%, reflecting improved clarity and convergence of opinion. Eight coders independently coded each comment based on the coding scheme shown in 1 and finalized an item’s code when 5+ authors agreed. Only one item did not reach consensus among the researchers (2, Item 96), but had already reached consensus among participants. Two coders independently reviewed 151 comments for the items that reached

Table 1: Coding scheme for Round 2 qualitative analysis.

Category	Description and Example	Number of Comments
Confusion	Expressed confusion about specific item and did not understand the phrasing (e.g., Conflict of Interest Statement (Item 10): “ <i>Do you mean stakeholders in the study?</i> ”).	10
“Only If”	Feedback was provided on items already labeled “Only if applicable to the study.” Since this feedback was already addressed by the existing instructions, it required no further action.	27
Placement	Comments on the positioning of some items, suggesting they fit better under another section. (e.g., Funding Statement (Item 9): “ <i>This could be given after peer review</i> ”).	2
Support	Rating supplements with additional feedback (e.g., Author Institutions (Item 3): “ <i>Honestly, it makes [authors] easier to find if you (...) do not have (...) a working email</i> ”).	90
Confusion/Support	Had initial confusion about the phrasing but later reflected and expressed support (e.g., Authors Names (Item 2): “ <i>but before what? You mean on a publication?</i> ”).	
Suggestion	Suggested modifications like placement, adding reasons, or rephrasing items (e.g., Sex and Gender (Item 28): “ <i>Authors should report how they define the sex/gender (...) for instance, whether they asked participants to self-identify or whether they used other sources.</i> ”).	5
Support and Suggestion	Supported their vote as well as suggested other ideas.	15
I Don’t Know	Some coders could not interpret the feedback. This prevented agreement in the category.	0

consensus, and afterwards discussed them. For the analysis and final revisions to the checklist, we considered only the following categories: Confusion, Suggestion, Support, and Suggestion. The resulting changes to the final checklist included rewording items (e.g., Item 1, 2, and 3: [After Peer review] to [Upon Publication]), adding clearer explanations (e.g., Item 30: criteria for needing sex/gender), and providing recommendations within the explanations (e.g., Item 67: number and duration of breaks, within and between sessions.).

Round 2 showed an increase in consensus (82%) compared to Round 1 (55%). The supplementary file contains a comprehensive report on participant consensus from the quantitative analysis of both rounds (Consensus Report, Appendix A).

5.3 Final Team Discussion

After the two Delphi rounds, all authors reviewed and discussed the findings. Out of the 114 items, 64 reached consensus in Round 1 and 50 in Round 2. Seven items were removed because they already encompassed within a combined category. For example, Item 17 contains both contributions and significance and Item 7 contains the abstract with all five components (Background, Purpose, Method, Results, and Conclusion). The team merged two overlapping items: Material/Equipment and External Recording Devices, as recording devices, such as a camera or a microphone, are considered a part of the Material/Equipment category. In addition, the team recommended adding the qualifier “Only if applicable to the study” to a few more items that did not already include it (e.g., Item 81: Interviews, as some HRI user studies do not involve Interviews). In total, we compiled the checklist with 106 items as shown in Table 2.

5.4 Guidelines

The goal of the RUSH checklist is to act as a supporting framework to bring rigor, transparency, and reproducibility in reporting HRI user studies. While developed and validated specifically for HRI user studies, the checklist can also support planning, execution/tracking, and reporting for work that includes a user evaluation with human

participants, in both controlled (e.g., lab settings) and uncontrolled (e.g., public settings) studies. Researchers are encouraged to begin using the checklist from study design through completion, updating it throughout the research process to ensure essential components are considered at all stages. Not all checklist items apply to every HRI user study. Researchers should mark non-applicable items and briefly justify any omissions. The checklist also aids reviewers and editors in evaluating the completeness and clarity of a study.

6 Discussion

The Delphi process generated a strong expert consensus on 106 reporting items across nine domains, forming the RUSH checklist: Reporting User Studies in HRI (2). A printable version of the checklist is provided as supplementary material, with complementary information for practical use (RUSH Checklist, Appendix B). To the best of our knowledge, RUSH is the first validated community-driven consensus-based reporting checklist for HRI user studies. Its successful adoption will require dedicated promotion through workshops, collaborative initiatives, and community engagement to ensure visibility, particularly among emerging researchers. Sharing this checklist and the work of 34 HRI experts (plus the authors), is just the beginning. We now discuss its implications and next steps for supporting rigorous HRI research within the community.

Community Uptake: The consensus achieved on reporting HRI studies, represented by the RUSH checklist reflects a shared foundation among HRI scholars, but its impact depends on its adoption. Other fields have achieved widespread uptake by integrating guidelines into journals and conference requirements [2, 30]. For example, NeurIPS² introduced a mandatory reproducibility checklist for paper submission in 2021 [31], and CONSORT is widely used in medical sciences after being endorsed by journals to improve transparency and quality in reporting randomized controlled trials [2]. For HRI, partnerships with leading venues, such as the HRI conference and major robotics journals, will be essential to

²<https://neurips.cc/public/guides/PaperChecklist>

Table 2: Final RUSH checklist, grouped by nine domains, showing the number of respondents (N) and the agreement percentage (%). Items marked (⊕) refer to “only if applicable to the study,” (Δ) refer to “Upon Publication,” and (★) refer to “Available By Request/Permission.” All items reached consensus: Round 1 (•; N=34) and Round 2 (•; N=28). Items with $\geq 80\%$ consensus are marked in blue; those with $< 80\%$ in orange.

A. General Information		
1 Title •	33, 91%	
2 Author Names • Δ	27, 96%	
3 Author Institutions • Δ	28, 89%	
4 City of Research • Δ	27, 89%	
5 Country of Research • Δ	34, 71%	
6 Study Dates (Start/End) • ⊕	28, 71%	
7 Abstract (Background, Purpose, Method, Results, Conclusion) •	28, 93%	
8 Study Preregistration Link ⊕ •	26, 77%	
9 Funding Statement • Δ	28, 100%	
10 Conflict of Interest Statement • Δ	28, 93%	
11 Author Contributions Statement •	27, 74%	
D. Study Design		
44 Equipment and Material • Robot	34, 85%	
45 Total No. of Robot(s) • ⊕ For Each Robot	34, 79%	
46 Type of Robot •	34, 88%	
47 Robot Model •	34, 76%	
48 Level of Autonomy •	34, 88%	
49 Reason for Using Robot •	28, 86%	
50 Commercial or Built for study •	34, 71%	
51 Picture of Robot •	34, 71%	
52 Type of Interaction •	34, 85%	
53 Communication Modality •	34, 91%	
54 Robot Location •	34, 82%	
55 Study Design (Between/Within-Subjects) • ⊕ Conditions	34, 94%	
56 No. of Experimental Conditions • ⊕	34, 94%	
57 No. of Participants per Condition • ⊕	34, 97%	
Participant Training Details		
58 Training Procedure • ⊕	34, 88%	
59 Training Duration • ⊕	34, 85%	
60 Training Stopping Criteria • ⊕	34, 82%	
61 Study Procedure •	34, 91%	
62 Interaction Method •	34, 94%	
63 Study Setup • ⊕	34, 88%	
Study Duration		
64 Full Session Duration •	34, 71%	
65 Duration of Participant-Robot Interaction • ⊕	34, 76%	
66 Interactions per Week • ⊕	28, 89%	
67 No. of Breaks • ⊕	28, 86%	
B. Introduction and Background		
12 Background/Related Work •	34, 74%	
13 Motivation •	34, 82%	
14 Objectives •	34, 88%	
15 Research Question/s •	34, 85%	
16 Hypotheses ⊕ •	28, 89%	
17 Significance and Contributions • (individually, Significance: 24, 85.7%; Contributions: 27, 96.4%)	25, 72%	
18 Long-term Goals ⊕ •	28, 89%	
E. Ethics		
68 Review Board Name ⊕ •	27, 85%	
69 Ethical Approval Statement • & No. ⊕	25, 88%	
70 Informed Consent •	34, 78%	
71 Data Protection & Privacy • ⊕	22, 82%	
72 Deception/Partial Disclosure • ⊕	34, 74%	
73 Debriefing ⊕ •	23, 87%	
F. Data Collection		
74 Additional Synthesis of Recordings to Collect Data • ⊕	34, 82%	
75 Sensors Used on Participants • ⊕	34, 79%	
Questionnaire		
76 Standard Questionnaires • ⊕	34, 82%	
77 Customized Questionnaires • ⊕	34, 76%	
78 Modifications to Response Options in Standard Questionnaires • ⊕	34, 82%	
79 Modifications to Items in Standard Questionnaires •	34, 82%	
80 Interviews • ⊕	34, 85%	
81 Observations • ⊕	34, 82%	
82 Analysis Framework •	34, 85%	
83 Data Processing •	34, 76%	
84 Coding • ⊕	34, 88%	
Statistics		
85 Use of NHST (Null Hypothesis Statistical Test) • ⊕	33, 76%	
86 Descriptive Statistics ⊕ •	34, 85%	
87 Specific Statistical Procedures ⊕ •	34, 82%	
Variables		
88 Independent Variables • ⊕	34, 85%	
89 Dependent Variables • ⊕	34, 88%	
90 Control Variables • ⊕	34, 85%	
H. Discussion		
97 Answer Research Question(s) •	34, 85%	
98 Limitations •	34, 79%	
99 Generalizability •	28, 79%	
100 Interpretation of Results •	34, 82%	
101 Future Work •	27, 81%	
C. Participants		
Population		
19 Intended Population •	34, 76%	
20 Actual Population • ⊕	34, 76%	
21 Total Recruited Participants •	34, 94%	
22 Sample Size Justification •	27, 96%	
23 Final No. of Participants •	34, 91%	
Dropouts		
24 Voluntary Dropouts •	28, 93%	
25 Dropouts Due to Pre-Screening •	34, 90%	
26 Dropouts Due to Study Interruption	27, 96%	
27 No. of Participants Removed from the Final Analysis •	34, 79%	
Sex and Gender		
28 Distinguish Sex/Gender ⊕	26, 81%	
29 Allow Multiple Gender Choices and Open Text Option ⊕	27, 81%	
30 Report Each Gender Category ⊕	27, 85%	
31 Include “Prefer not to disclose” in Gender Categories ⊕ •	34, 76%	
32 Age • ⊕	34, 79%	
33 Education ⊕ •	28, 89%	
34 Occupation ⊕ •	28, 82%	
35 Technology Familiarity ⊕ •	28, 93%	
36 Prior Robot Experience ⊕ •	28, 93%	
37 Disability ⊕ •	28, 82%	
38 Inclusion and Exclusion Criteria with Justification • ⊕	34, 74%	
39 Recruitment Method •	34, 76%	
40 Setting •	34, 88%	
41 Compensation •	34, 71%	
Consent		
42 Consent Format ⊕ •	27, 89%	
43 Consent Provider ⊕ •	27, 85%	
G. Results		
91 Findings •	34, 90%	
92 Full Description of Analysis •	28, 93%	
93 Significant Results ⊕ •	34, 91%	
94 Non-Significant Results ⊕ •	28, 93%	
95 Statistical Reporting ⊕ (Degrees of Freedom, Test Statistics, Effect Sizes, Confidence Intervals, and p-values) •	28, 100%	
96 Descriptive Analysis ⊕ •	34, 71%	
I. Code and Data Release		
102 Code Availability • ★	27, 89%	
103 Repositories or Notebooks • ★	26, 81%	
104 Data Collection Scripts • ★	28, 75%	
105 Data Cleaning & Analysis Scripts • ★	28, 82%	
106 Online Datasets • ★	27, 96%	

embed reporting practices, which could be incentivized via required checklist completion at submission, by providing reviewers with the checklist as a review aid, or integration into paper templates.

Beyond Checklists - Templates, Training, & Community Support: Checklists alone are insufficient. Their effectiveness relies on shared understanding and consistent use. Evidence from clinical sciences shows that even widely adopted guidelines (e.g., CONSORT) benefit from supporting resources and training, such as those provided by the EQUATOR Network³. In HRI, this could include explanatory materials, annotated examples, workshops, and integration into graduate curricula and community support, such as an HRI helpline or Slack mentoring channel to provide early guidance for study planning, and improve quality, consistency, and ethical rigor. Space limitations in conference submissions often force tradeoffs in reporting. Structured checklists with concise explanations offer a practical way to maintain transparency. Tailored checklist versions can further assist: a condensed “essentials-only” checklist for short submissions (4-6 pages) and the full version for longer conference papers (8-10 pages) and journal articles. We strongly recommend integrating the checklist into custom Overleaf or Word templates for top-tier venues like the HRI conference. This ensures consistent use, reduces time, effort, and human error (e.g., omitting items), and eliminates the need for separate documents, including our checklist or forms like that proposed by Holthaus et al. [17]. We plan to request a trial run at next year’s HRI conference, which would support normal reporting practices while enabling collection of community feedback on its usability.

Living Document and Periodic Revisions: Reporting guidelines cannot be static documents. As research practices evolve, checklists must be periodically updated. Prevalent guidelines such as NeurIPS Reproducibility Checklist [31], ACM artefact Review and Badging⁴, CONSORT [18], PRISMA [30] are periodically revised by steering groups and publish updates and extensions. Similarly, the RUSH checklist should remain a living document, with revisions informed by community feedback, emerging technologies, and study designs. To support sustainability, a dedicated steering group could oversee revisions at regular intervals (e.g., every 5 years) or in response to major methodological developments, supported by an open platform for ongoing community feedback.

Implications for HRI: Reporting guidelines have been transformative in fields such as clinical trials, where CONSORT improved transparency, comparability, and research quality [25]. Similarly, SPIRIT (Standard Protocol Items: Recommendations for Interventional Trials) defined essential protocols, enhancing transparency, consistency, and quality in trial design and reporting [26]. In Computer Science, the NeurIPS Reproducibility Checklist standardizes the reporting of experimental details, improving transparency, peer review, replication, openness and accountability [31]. Similarly, in HRI, our checklist can enhance transparency in study design, execution, and analysis, paving the way for systematic reviews and meta-analyses. Standardized reporting can also prevent redundancy, support cumulative knowledge building, and strengthen the evidence base for both theory and application.

Ethical and Impartial Reporting: The RUSH checklist emphasizes transparent reporting across all sections of HRI user studies to ensure replicability and integrity. Ethical reporting also requires documenting explicitly informed consent, ethics approval, participant selection criteria, and potential risks and mitigation strategies. As reported by Seaborn et al., consistent demographic reporting can help evaluate robotic systems across diverse populations, reduce biases and promote equitable design [38]. These considerations, however, extend beyond demographics: researchers should also consider accessibility, societal and geographic representation, and recruitment practices that may systematically benefit or exclude individuals. To support this, the RUSH checklist could include items on demographics, ethics approval, accessibility, and sample limitations, while ensuring participant privacy is protected.

6.1 Limitations & Future Work

Despite our best efforts for an open and broad-reaching community process, we acknowledge several limitations. First, we relied on our networks and knowledge of major players in HRI to recruit for the study. This could have excluded viable recruits. Second, we conducted the study in English, the language currently recognized as global. This could have excluded certain parties or made participation difficult. Finally, we recognize the additional burden a checklist like this can impose on practice. The best way forward may be to work with a standards body and certain venues like the HRI conference to integrate the framework or develop a bare-minimum version during publication cycles. The viability of this should be explored in community venues like workshops and town halls.

Future work should examine the adoption of the RUSH checklist across HRI venues and identify barriers to consistent use. Linking the checklist to pre-registration platforms (e.g., Open Science Foundation) and open data repositories could further promote transparency and streamline reporting. Longitudinal studies incorporating the RUSH checklist will be essential to assess the checklist impact on methodological rigor and reproducibility in HRI research.

7 Conclusion

We developed and validated RUSH, a new consensus-based tool for reporting HRI studies informed by 34 HRI experts from around the world. Over two Delphi rounds, experts provided feedback that enabled consensus about items that should be openly reported when describing HRI research. We offer it to the community through an open platform: <https://aspirelab.io/rush>. As a community, our collective next step is to take up this tool and apply it to our work. Years from now, we will be able to assess its efficacy at enabling transparent reporting, clarity in descriptions of HRI studies, and meta-analyses for the sake of knowledge production.

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³<https://www.equator-network.org>

⁴<https://www.acm.org/publications/policies/artifact-review-and-badging-current>

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